



# बड़हवा RACING

Newsletter - April 2016







## Newsletter - April 2016



Dear Readers,

The past 10 months building the race-cars have been fun and challenging for the team. With the launch of our combustion and hybrid prototypes, we aim to conquer new frontiers. The hybrid team will be participating in the Formula Hybrid Event at New Hampshire from 2<sup>nd</sup> to 5<sup>th</sup> May, 2016. We aim at securing a position among the top 5 teams in the world.

The testing phase of our latest combustion prototype is now underway. Over the next 4 months, the car will be pushed to its limits and checked for its reliability and performance. The combustion team will be participating in FS Czech Republic which is to be held from the 3<sup>rd</sup> to 6<sup>th</sup> of August, 2016.

In this edition, we draw attention to the launch event and the team's activities over the past 3 months. The support provided by our sponsors has played an important role in prototyping. Also, the encouragement from our College and Rashtreeya Sikshana Samithi Trust, Our College faculty and our beloved Alumni has been instrumental towards the project's accomplishments.



## 2016 Rollout



*“Hope  
Smiles from the threshold of the year to come,  
Whispering 'it will be happier'...”*  
— Alfred Lord Tennyson

Ashwa Racing's 12<sup>th</sup> race-season kicked off with the roll-out of the 2016 prototypes, the RZX6-C and RZX6-H. The launch event was held on the 27<sup>th</sup> of February, 2016. It was a delight to be host to our Sponsors, Trust and College representatives, our alumni, parents, well-wishers and everyone present at the event. It was a great pleasure speaking about our journey over the past 10 months.

We would like to thank everyone who has been a part of our project and it is this relationship that we shared, which we will carry with us throughout our life.





Launch Presentation



Cars unveiled by  
our sponsors



Our Sponsors with  
Our cars





# ashwath RACING

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Our 2016 Com-  
bustion car, The  
RZX6-C, ready to  
zoom



The RZX6-H  
preparing for its  
maiden run

The prototypes after a  
busy day







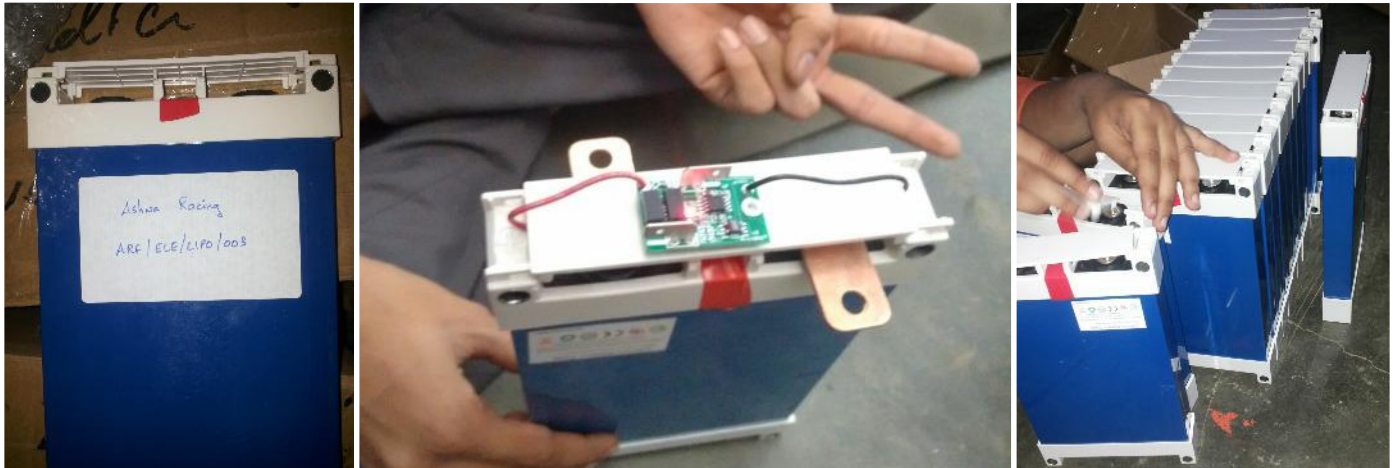
Recollection  
of few of the  
joyful  
moments that  
we  
will cherish  
through-out our  
lives.







## On the Electric Front



### Competition approved battery packs with custom BMS

The most important component procured this race season were the Lithium Ion batteries for the Hybrid. The previous battery packs had internal wiring issues which stood in our way to clear the Hybrid scrutiny.

After several failed attempts to re-wire the previous batteries, the team decided to procure better, competition approved battery packs. Our electric sub-system mentor Mr. Nishant Kalbhor, directed us towards Liyuan battery packs and we subsequently got them approved by the event organizers. Mr. Kalbhor also helped us procure these batteries at a discounted rate and they reached our workshop in January.

Upon arrival of the packs, the final wiring of the Hybrid was initiated. Fuses and other required components were procured. Fire retardant UL Grade battery boxes were manufactured at Aradhana Fibreglass Technologies. Fuses and several devices were procured. Controllwell India Pvt. Ltd. extensively helped by providing us with UL listed Conduits and Connectors. Copprrod Industries Pvt. Ltd. a country leader in Bus Bar manufacturing, helped us with the required tin coated Bus Bars. Our close sponsor since many years, ABB India Limited, provided us with the required High Voltage Disconnect Switch and necessary sensors. The High Voltage wires were crimped at LAPP INDIA and our battery charger reached us in February. We thank Meritor for the help extended to ship our components to India.

We would also like to mention our 2010 team member, Mr. Nikhil Jali, who currently works at Purdue University, has helped us immensely in mastering the Hybrid architecture.



## On the Electric Front



Car during Electrical Assembly



Elcon Tech  
programmable  
charger

High Voltage Discon-  
nect Switch from ABB

## Kalbhorz Electric Support

Mr. Nishant Kalbhor, Founder of Kalbhorz Electric, has been our mentor and technical advisor for the Hybrid car. Kalbhorz Electric also helps our team by providing Motors.

During his time in college, Mr. Nishant entered the world's first electric bike race at the TTXGP, Isle of Man, and secured the 3<sup>rd</sup> position. The team was called Tork India.

The next year, Team Tork competed in the world's first Electric Bike championship and won the first race, beating even higher powered vehicles. This was credited to the design strategy of achieving the best possible power to weight ratio.

Team Tork then contacted several companies to build bikes. They also entered in 2012 for the Isle of Man. It was here that the need for innovative custom parts was realized.

Mid 2012, Kalbhorz Electric was formed to develop indigenous Motors controllers and Battery Management systems. With three years of R&D in electrical vehicle architecture, Kalbhorz Electric is now a proud company as a supplier to some of the biggest names in India.





## Indo-German Chamber of Commerce and Ashwa Racing



The relationship that Ashwa Racing shares with IGCC, Bangalore has proven to be invaluable to the team. Thank you, IGCC for all the support.

Ashwa Racing at the IGCC Christmas party. Had a wonderful time.



RZX6-C at the IGCC Football Tournament. We congratulate BOSCH India on winning the tournament.





## Bodywork Manufacturing at FIE TECH

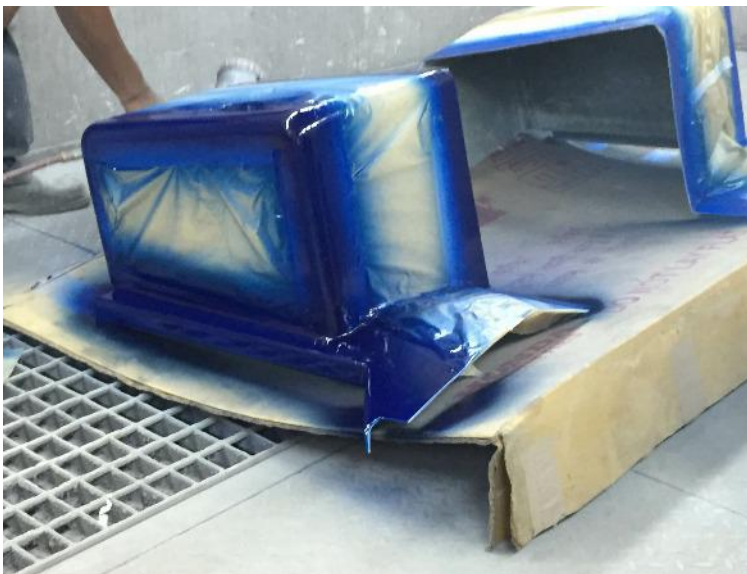


The bodywork for the 2016 cars were manufactured at FIE TECH, a specialist in the manufacture of fiberglass components and structures. FIE TECH has been supporting us with beautifully crafted lightweight body over the past two years.



**FIE TECH**  
*Fantasy in Fibreglass*

## Bodywork Painting at Magnum Honda



We would specially like to mention and thank Magnum Honda for painting the bodywork. Magnum Honda is South India's No.1 Honda car dealer.







## CMM of Chassis



The 2016 chassis were tested for conformance with designs with the help of portable CMM arm offered by Nebula Technologies. Subsequently, the necessary changes will be made to the fixtures for our next year's cars to reduce the errors. We thank Nebula Technologies for the support.

## ACMA and Ashwa Racing

**ACMA**  
Automotive Component Manufacturers Association of India  
(Southern Region – Karnataka & Hosur)

**Mr Ram Venkataramani, Chairman & The Regional Committee**  
ACMA Southern Region

*Request the pleasure of your company to meet and interact with*

**Mr Arvind Balaji, President & Mr Rattan Kapur, Vice President**

**Chief Guest: Shri. R.V. Deshpande – Minister for Industries Govt. of Karnataka – Large & Medium Enterprises & Tourism**

*On Monday, 21<sup>st</sup> March 2016*

We were invited for an interaction conducted by ACMA on the 21<sup>st</sup> of March, 2016. Various member companies were present for the interaction and the minister of tourism and industry, Karnataka, Mr. R.V. Deshpande, was

the chief guest. Apart from highlighting a few government policies, the minister also stressed on the importance of Research and Development. After this, the team got an opportunity to interact with representatives from many automotive manufacturers.

We thank Ms. Neeraja Raj from ACMA for inviting us to the event. It is opportunities such as these which help us reach out to potential partners.

Being a student R&D project, and keeping in mind the minister's emphasis on research, we hope that automotive component manufacturers come forward to support our endeavour.





## New faces in the team



Recruitment of first year members into Ashwa was initiated in the month of September, 2015. With the assistance of the HR team, our alumnus, Mr. Siddharth Mandal presided over this exercise. The process began with an aptitude test where 150 out of 250 people were selected. This was followed by 2 rounds of personal interviews conducted by the HR team and by our founder, Mr. Arunraj, respectively. A group of 40 students were chosen to undergo training during the winter holidays.

To make this activity enjoyable, many challenges were presented to the freshers. One such challenge was the "cart event", where people were divided into teams and were asked to build a single seat cart. The teams were evaluated on their creativity, their basic technical knowledge, and their ability to manage resources and market their product. This crude emulation of "Formula student" has also proven to be an effective team building exercise over the past years. Finally, 30 members were selected.





## Happy Journey RZX6-H



Our Hybrid Packed and ready to leave for the long journey to USA

After the launch event on the 27<sup>th</sup> of February, the Hybrid was tested for the next two weeks.

The initial plan was to ship the car on the 20<sup>th</sup> of March. However, as our cargo was delayed at the US customs last year, our logistics partner advised us to ship the car 10 days in advance.

After packing all necessary items and spares, the car was packed into the cargo box, which was built by Bhumiputra Architecture.

The cargo left the workshop on the 10<sup>th</sup> of March, 2016. The car is scheduled to arrive at the New Hampshire Motor Speedway by the 26<sup>th</sup> of April.

We are now preparing for the Static events. The 2016 combustion car is being tested and the team has also begun its designs for the 2017 race season.



The cargo leaves workshop late evening on the 10<sup>th</sup> of March





## Alumni Focus



Uday and RZX5-C

Our 2015 batch team member Mr. Uday Kumar, who currently works as a Graduate Engineering Trainee in TATA Consulting Engineers Limited, helped our team in the time of utmost need. We would like to thank Uday for being our strength and holding our team together during dire times and during times of internal conflict.

Uday was the Head of Chassis and Manufacturing during his tenure in Ashwa Racing. Thank you Boss for your guidance. Without you we would have crumbled.



Saif Absar at Formula Hybrid 2015

We would also like to thank our 2015 batch team member, Mr. Saif Absar for helping us with various problems that we faced in the making of our 2016 Hybrid car.

Saif currently works at Ather Energy as an application engineer.

Saif worked on the 2015 Hybrid car electrical architecture when in Ashwa Racing. He continues to work with us closely to improve the hybrid architecture. Saif has also helped us by providing the Battery Charger in times of need, to charge our Lipo Batteries.





## Sponsor Focus



Schneider Electric, our sponsor since our very first Demo Hybrid car, continues to support us and we are forever grateful for the encouragement towards our project.

Schneider Electric SE is a French multinational corporation that specializes in electricity distribution, automation management and produces installation components for energy management.

Ace Manufacturing Systems Ltd., is one of the largest CNC machine tool builders in India today. The company has expertise in manufacturing of CNC vertical machining centers, CNC horizontal machining centers and in providing manufacturing solutions centered around these products.

Ace Manufacturing Systems Ltd. Joined our family of sponsors in 2015 and continues to support us and put our efforts on the global platform.



Infineon Technologies AG is a world leader in semiconductor solutions that make life easier, safer and greener. Microelectronics from Infineon is the key to a better future. With products for the Automotive sector, Industry power control, power management and chip security sectors, Infineon Technologies continues to rule the semiconductor markets.

We welcome Infineon Technologies to our family of sponsors. THANK YOU SIR 😊.

Ajax-Fiori is a construction machinery manufacturer which offers a ranges of products that are Self-Loading Concrete Mixers, Concrete Batching Plants, Site Dumpers, Tunnel Dumpers and Concrete Pumps, being used across varied application sectors, viz., Irrigation, Infrastructure, Urban Development, Power Housing, Industrial Projects.

We welcome you to our family and thank you for your support.







# ashwa RACING

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**SPICER®**  
Drivetrain Products

Spicer India Private Limited is a leading manufacturer of Axles for light vehicles and drive shaft (propeller shaft) assemblies for a wide range of vehicle applications such as on and off-highway vehicles, industrial machinery and commercial vehicles that offer the multifarious advantages of improved fuel economy, torque-carrying capacity, weight reduction.

We thank Spicer India for believing in us and supporting us at the time of need. Spicer India supports us both with finances as well as with the production of shafts.

SABIC is a Saudi Arabian diversified manufacturing company, active in chemicals and intermediates, industrial polymers, fertilizers, and metals. It is the largest public company in Saudi Arabia. SABIC is also the largest listed company in the Middle East and one of the world's largest petrochemical manufacturer.

We welcome SABIC to our family of sponsors. SABIC has whole heartedly supported us and believed in our project. We are ever grateful for the support.

سابك  
sabic



Meritor Inc. is an American corporation headquartered in Troy, Michigan, which manufactures automobile components. It is a Fortune 500 company.

Meritor Commercial Vehicle Systems India Pvt. Ltd., the Indian subsidiary, is one of our strongest pillars of support for both combustion and hybrid prototypes. Their support has been instrumental in the making and the running of the hybrid car. We would like to thank Mr. Pradeep Mirji, Mr. Giriprasad and Ms. Soumya for their immense support towards the team.

HARTING India is a 100 % subsidiary of HARTING Technology Group - Germany. HARTING Technology Group has 4,200 people across its 43 sales companies and 13 production plants worldwide. The HARTING range of products and solutions comprises of connectors, device connection technology and network components. HARTING products connect machines and facilities via data, signal and power. These offerings enable solutions for the following markets: automation technology, energy, transport technology, radio, stage and event management technology machine construction, medical technology and infrastructure.

We whole heartedly welcome HARTING India to the Ashwa family. We thank you for your support extended in the time of utmost need.



**Pushing Performance**



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Atlas Copco (India) Ltd. started its operations in 1960, and has 22 offices across India, the registered office being at Pune. In 2015, Atlas Copco (India). Their products and services range from compressed air and gas equipment, construction and mining equipment, industrial tools and assembly systems, to related aftermarket and rental.

Atlas Copco provides its customers with the solutions they need to enhance productivity and achieve their manufacturing goals.

We whole heartedly welcome Atlas Copco to the Ashwa family. Thank you for your support.

FIE TECH is a specialist in manufacture of Fibreglass components and structures. FIE TECH's specialty lies in in-house Design & Development of all the Products and Manufacturing the same with high quality keeping in view of aesthetics and purpose. FIE TECH primarily takes up Turnkey Projects of Water Parks, Theme Parks, Art Works & Interior for Hotels, Public spaces, Offices, Buildings.

We thank FIE TECH for the immense continued support towards our project. Mr. Vinay Pawar, MD, Fie Tech, has always whole heartedly, gone out of his way to help us. Thank you sir.



**FIÈ TECH**  
*Fantasy in Fibreglass*



TE Connectivity (TE), is a global company and a industry leader in highly engineered connectors and sensors. TE designs and manufactures connectivity and sensor solutions for a variety of industries including automotive, industrial equipment, data communication systems, aerospace, defense, oil and gas, consumer electronics, energy and subsea communications. The company serves customers in more than 150 countries.

We thank TE Connectivity for the continued support and also for providing us with finances for this race season.

Controlwell India Pvt. Ltd. Is a leading distributor of quality Cable Protection and Management products in India. Controllwell helps us by providing imported certified conduits, connectors, junction boxes, grommets and many more High Voltage Accessories.

We would specially like to mention and thank Ms. Neelu Saheta, CEO, Controlwell India Pvt. Ltd. We are forever thankful to you for all the support you have provided, especially when asked at very short notices. Thank you.

**CONTROLWELL**





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## MOTUL



Motul is a French company, now worldwide, which creates, develops and distributes lubricants for engines (motorcycles, cars and other vehicles) and for the industry. Today, Motul is present in more than 80 countries and designs, elaborates and distributes lubricants with higher technical value added. As a pioneer in many synthetic and semi-synthetic products, Motul has always favored innovation, research and development.

We thank them for the vast range of products that they have provided us.

## element14

Element14, is a distributor of technology products and solutions for electronic system design, maintenance and repair. It is the Asian Pacific trading brand of global electronics distributor Premier Farnell plc, which is a constituent of the FTSE 250 Index listed on the London Stock Exchange.

We thank Element 14 for all the components provided.





ABB is a Swiss multinational corporation headquartered in Zurich, Switzerland, operating mainly in robotics and the power and automation technology areas. It ranked 158th in the Forbes Ranking.

We would like to extend our hearty thankfulness towards Mr. Dev Chandra Kuril of ABB for supporting and helping us with components such as the current sensor and the High Voltage Disconnect (HVD), both which play a key role in the working and safety of the hybrid architecture.



Stumpp, Schuele & Somappa are a part of the MG Brothers Group, a 60 year old enterprise which has various businesses such as Transportation, Auto Dealerships, Agriculture & Real Estate to name a few.

With a capacity to convert 15,000 tons of steel into springs each year; Stumpp, Schuele & Somappa is the "Number 1" spring maker of high quality springs in India. Established in 1960, Stumpp, Schuele and Somappa was the first spring manufacturing unit in the sub continent.

Stumpp, Schuele & Somappa Group has been our sponsor for all our spring requirements from the very beginning. We thank Stumpp, Schuele & Somappa Group for the support towards our project.

FESTO is a German industrial control and automation company based in Esslingen am Neckar Germany. FESTO is an engineering-driven company, producing and selling pneumatic and electrical control and drive technology for factory or process automation. FESTO Didactic is a world market leader in industrial education and consultancy. Sales subsidiaries, distribution centers and factories are located in 61 countries worldwide.

We thank FESTO for the support towards our project. Their support has helped us implement various pneumatic systems in our car.



Suprajit is a customer centric company taking on challenging product development while manufacturing to international specifications for a wide range of control cables and instruments.

We thank them for the cables that they have provided.

Aradhana Fibreglass Technologies is one of the illustrious manufacturers and developers of FRP products such as FRP sheets, FRP roofing, FRP insulation and automobile components, etc.

AFT has helped us with the manufacture of Fire Retardant Battery Box and components that are compliant with international standards.

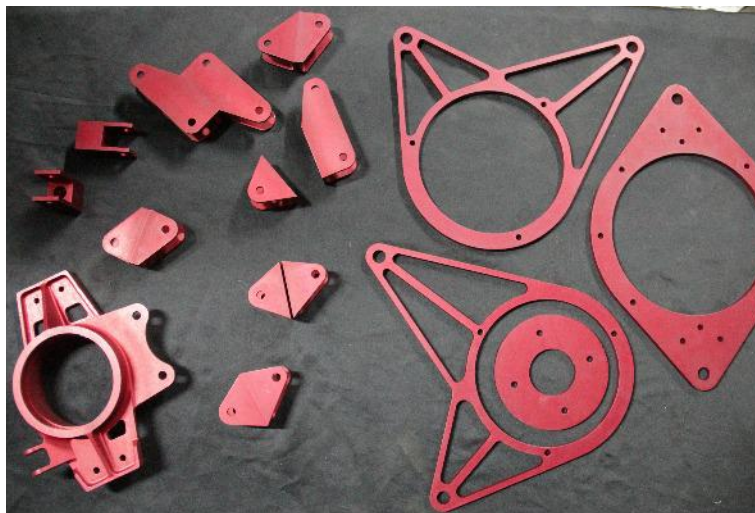






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G. V. Enterprises, are one of the premier industries in the field of anodizing and fabrication of Solar Module Frames since 1992. They are specialized in the field of anodizing and electro coloring of all profiles.

We thank them for anodizing our components in beautiful colors.



www.smartlinkinc.com

SmartLink provides complete project solutions. We thank them for providing us with fasteners for our prototypes.

Lore Software Solutions, a Bangalore, India based IT services company, provides software development, quality assurance, product certification and operations management services and specializes in the BFSI and Content Management domains.

We thank them for supporting us with finances for the 2016 race season.



restaurant | caterers | campus café

We would like to thank SRV Solutions and OGARA Campus Cafe for joining us our launch event and catering sponsor for our launch event of 2016 prototypes.

Miranda Tools manufactures HSS Tools Bits, Drills, End mills, Reamers, HSS Saws and Metal Cutting Bandsaw Blades. It has its manufacturing unit & marketing office at Ankleshwar in Gujarat in western India . It is equipped with state of the art technology for producing precision cutting tools confirming to IS, BS, DIN, JIS & ISO standards.

We thank them for the drill bits that they have provided us.





Receipts and payments Accounts for the period from 26-02-2016 to 27-03-2016

| RECIEPTS                                            | AMOUNT           | PAYMENTS                                                                                                                                                               |
|-----------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opening Balance in<br>ING Vysya Bank<br>RVCE Branch |                  | By Chassis and Suspension :Rs. 10,339.00<br>PT & SMP :Rs. 17,080.00<br>Electrical :Rs. 5,981.56<br>Brakes & Logistics :Rs.5,32,145.20<br>Miscellaneous :Rs.1,35,962.00 |
| SB A/c 136010060793                                 | :Rs.8,60,903.11  |                                                                                                                                                                        |
| SB A/c 136010058514                                 | :Rs 2,94,740.26  |                                                                                                                                                                        |
| SB A/c 136010046665                                 | :Rs 8,453.51     |                                                                                                                                                                        |
| Cash in Hand                                        | :Rs 6,441.80     | Bank Charges:<br>SMS and Wire Transfer :Rs 72.98<br>Cash in Hand :Rs 91,115.08                                                                                         |
| Sponsorship deposit                                 | :Rs. 4,50,000.00 |                                                                                                                                                                        |
| Deposit by 2016 Team-<br>VISA processing fee        | :Rs. 2,35,200.00 | Closing Balance in<br>ING Vysya Bank<br>RVCE Branch                                                                                                                    |
| Deposit by 2017 team-<br>seed money                 | :Rs. 1,04,024.00 | SB A/c 136010060793 :Rs.6,18,272.11<br>SB A/c 136010058514 :Rs.2,25,506.54<br>SB A/c 136010046665 :Rs.4,58,298.25                                                      |
| Deposit by 2016 Team-<br>USA accommodation          | :Rs. 25,000.00   |                                                                                                                                                                        |
| FS Czech registration<br>deposit                    | :Rs. 1,10,000.00 |                                                                                                                                                                        |
| TOTAL :                                             | Rs. 20,97,464.1  | TOTAL : Rs. 20,97,464.1                                                                                                                                                |

Checked and Found Correct

Finance Team  
बड़ेबाRACING  
R.V. College of Engineering





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## OUR HYBRID SPONSORS







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